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Distributed hierarchical control of islanded active distribution networks with dynamic consensus algorithm and adaptive virtual impedance

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In the event of an upstream grid fault, the Active Distribution Network (ADN) can switch to autonomous operation, coordinating controllable Distributed Generators (DGs) and Energy Storage Systems (ESS) to maintain power supply reliability. To achieve full decentralization, improve reliability, and mitigate the adverse effects of feeder impedance mismatches, this paper proposes a distributed Hierarchical Control (HC) strategy that integrates adaptive virtual impedance with a dynamic consensus algorithm. The proposed scheme enables accurate power sharing and simultaneous restoration of system frequency and average voltage—without requiring prior knowledge of grid impedance parameters. The feasibility of the proposed control scheme is validated through multi-scenario case tests and real-time simulation based on OPAL-RT 5707 platform.

KEYWORDS

active distribution network, adaptive virtual impedance, dynamic consensus algorithm, hierarchical control, power sharing

1 Introduction

With the global transition towards a low-carbon energy system, the penetration of renewable energy in distribution networks is increasing [Jia et al. \(2015\)](#); [Zhang et al. \(2015\)](#). This shift is transforming traditional passive networks into interactive systems known as the Active Distribution Network (ADN). The core feature of ADN is its ability to flexibly coordinate and dispatch Distributed Energy Resources (DERs)—including Distributed Generators (DGs), Energy Storage System (ESS), and flexible loads—thereby significantly enhancing operational efficiency and reliability [Nick et al. \(2014\)](#); [Hasheminamin et al. \(2015\)](#). Notably, during upstream grid faults, ADNs can switch to islanded operation, using internal resources to maintain an uninterrupted power supply to critical loads [Wu et al. \(2014\)](#).

To improve power supply reliability, research on Energy Management System (EMS) architectures for coordinating DERs has become a major focus [Chegari et al. \(2024\)](#). Existing control architectures are primarily categorized as centralized, decentralized, or distributed [Zhao et al. \(2018\)](#). Centralized control relies on a Central Control Center (CCC) to gather global information for unified

optimization Zhou et al. (2017); Liu et al. (2023); Akarne et al. (2024). While capable of achieving a global optimum in small-scale systems with ideal communication, it suffers from the risk of a single point of failure, high communication costs, and a computational burden that grows exponentially with the number of DERs Mehdi et al. (2020). These limitations restrict its applicability in large-scale, dynamically changing ADN Elnady et al. (2022).

Decentralized control, such as conventional droop control, has emerged as an alternative Liu et al. (2022); Lai et al. (2019). In this scheme, each DG unit adjusts its output using only local measurements without communication with other units, making it suitable for areas where communication is constrained Xie et al. (2023). This grants the system high modularity, reliability, and plug-and-play capability Yang et al. (2024). However, due to differences in feeder impedances and unequal electrical distances to the Point of Common Coupling (PCC), traditional P-f/Q-V droop control often fails to achieve accurate reactive power sharing and introduces steady-state voltage and frequency deviations Liu et al. (2024).

To balance the global optimality of centralized control with the high reliability of decentralized control, Hierarchical Control (HC) architectures are widely adopted, making them well-suited for medium-to-large-scale smart distribution networks with numerous DGs Li et al. (2018). This architecture typically comprises three layers with different time scales: Primary Control (PC) provides a fast local response to establish voltage/frequency and achieve preliminary power sharing; Secondary Control (SC) acts on a slower timescale to eliminate steady-state deviations and correct sharing inaccuracies; Tertiary Control (TC) focuses on long-term economic optimization Shafiee et al. (2014); Li et al. (2022). However, when SC is implemented centrally, the fundamental issues of communication bottlenecks and single-point failure persist Shafiee et al. (2014).

Recently, with advances in Multi-Agent System (MAS) theory, distributed SC has gained significant attention Li et al. (2022). By forming a sparse communication network where agents only exchange information with neighbors, DERs can collaboratively restore system frequency and voltage Liu et al. (2020). This approach combines the coordination benefits of centralized control with the robustness of decentralized control. To mitigate the uneven power sharing caused by feeder impedance mismatches, researchers have developed impedance reshaping-based approaches, which can be broadly categorized into fixed virtual impedance (FVI) and adaptive virtual impedance (AVI) Deng et al. (2023). FVI typically superimposes a predefined fixed virtual impedance onto the output impedance of the inverter to mitigate the impact of line impedance mismatches on power sharing Qi et al. (2020), Mohammed et al. (2023). This method does not require complex algorithms and incurs low computational burden; however, its power sharing accuracy is limited, making it suitable only for scenarios with small line impedance differences and relatively stable loads. To achieve more accurate impedance compensation, a recursive least squares-based online impedance estimation algorithm was proposed in Mohammed et al. (2023), enabling zero-error power sharing. Nevertheless, this approach requires real-time measurement of voltage at the point of PCC and employs centralized communication to coordinate the output voltage and current information of each DG for impedance estimation and

voltage restoration, thereby failing to address the risk of single-point failures. In contrast, AVI dynamically adjusts the virtual impedance value according to real-time system conditions (e.g., output current, power deviation), enabling more precise power sharing and thus demonstrating superior performance in complex meshed networks and dynamic topologies Deng et al. (2023). In Vijay et al. (2021), the virtual impedance is designed as a function of the local output current of the inverter and integrated into the existing droop control framework. The major advantage of this method is that it relies solely on local information, eliminating the need for communication among DGs. However, it still suffers from certain power sharing errors, and the design of the functional relationship remains challenging. To improve dynamic stability, Alzayed et al. (2025) proposed a neural-network-based virtual impedance technique, which is particularly effective for highly nonlinear systems. Yet, minor steady-state deviations may occur under large impedance variations. In Xiao et al. (2024), adaptive virtual impedance effectively addressed the issue of active power and harmonic power sharing in low-voltage distribution networks, although reactive power sharing and voltage restoration were not considered. Reference Liu et al. (2025) proposed an AVI algorithm aimed at resolving the conflict between reactive power sharing and voltage restoration, but the entire process was implemented in an inductive network. Differently, Abouassy et al. (2025) simultaneously considered both active and reactive power sharing, where the virtual resistance is generated based on active power sharing error and the virtual reactance is derived from reactive power sharing error. Although this method accounts for the resistive component's influence, it does not consider the power coupling issues inherent in resistive-inductive networks.

Although the aforementioned studies have achieved significant progress, the coordinated design of distributed control strategies that simultaneously ensure accurate power sharing and high-quality frequency and voltage regulation—particularly in systems with large impedance mismatches and frequent load fluctuations—remains a challenging and worthy topic for further investigation Deng et al. (2023), Vijay et al. (2021), Xiao et al. (2024).

To address these challenges, this paper proposes a distributed hierarchical control strategy that integrates a Dynamic Consensus Algorithm (DCA) with AVI.

The main contributions of this paper are as follows:

1. A distributed cooperative control framework is designed, requiring only local communication among neighboring DGs, thereby eliminating dependence on a central controller. It simultaneously restores the system's average voltage and rated frequency while ensuring accurate active and reactive power sharing.
2. An online adaptive virtual impedance mechanism is embedded into the consensus algorithm. Without prior knowledge of network topology or line parameters, it dynamically compensates for feeder impedance mismatches, effectively solving the power coupling problem in resistive networks and improving control flexibility.
3. Multi-scenario case studies demonstrate that the proposed strategy maintains plug-and-play functionality and exhibits resilience against common disturbances such as communication failures and delays.

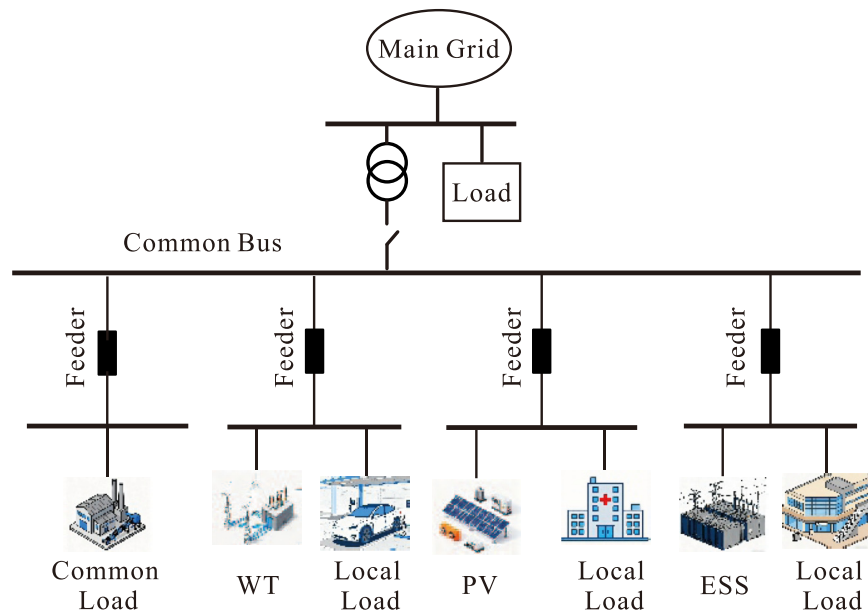


FIGURE 1
Schematic of an example ADN.

The remainder of this paper is organized as follows. Section 2 reviews the theoretical background of MAS communication graphs and the dynamic consensus algorithm, and analyzes the power-sharing problem. Section 3 details the proposed distributed hierarchical control strategy. Case studies are presented in Section 4, where the strategy is validated under various scenarios using MATLAB/Simulink. Section 5 further verifies the approach through real-time simulation on the OPAL-RT platform. Finally, Section 6 concludes the paper.

2 Theoretical preliminaries

2.1 Power sharing analysis

An ADN comprising DGs and loads is depicted in Figure 1. Each DG is connected to the common AC bus via its grid-tied power electronic inverter and respective feeder impedance. When an upstream grid fault occurs and the ADN is disconnected from the main grid and operates in islanded mode, the internal DGs coordinately supply power to critical loads. The active and reactive power injected from the i -th DG to the common bus can be expressed as:

$$\begin{cases} P_i = \frac{V_{oi}V_{pcc} \cos(\theta_i - \delta_i) - V_{pcc}^2 \cos \theta_i}{Z_i} \\ Q_i = \frac{V_{oi}V_{pcc} \sin(\theta_i - \delta_i) - V_{pcc}^2 \sin \theta_i}{Z_i} \end{cases} \quad (1)$$

where subscript “ i ” represents the parameters of the i -th DG, V_{oi} is the output voltage amplitude of DG, V_{pcc} is the common bus voltage amplitude, δ_i is the power angle between V_{oi} and V_{pcc} , Z_i and θ_i are

the magnitude and phase angle of the feeder impedance, respectively, and P_i and Q_i are the output active and reactive power, respectively.

Assuming a predominantly inductive feeder impedance ($\theta_i \approx 90^\circ$) and a small power angle during normal operation ($\delta_i \approx 0$), Equation 1 can be simplified to:

$$\begin{cases} P_i = \frac{V_{oi}V_{pcc}}{X_i} \delta_i \\ Q_i = \frac{V_{oi} - V_{pcc}}{X_i} V_{pcc} \end{cases} \quad (2)$$

where X_i is the imaginary part of Z_i .

In this case, the active power is strongly correlated with the power angle, while the reactive power is predominantly influenced by the voltage amplitude difference. This decoupling characteristic forms the basis for conventional droop control.

For decentralized power sharing, the conventional $P-\omega$ and $Q-V$ droop control is typically implemented locally for each DG:

$$\begin{cases} \omega_i = \omega_n - m_i P_i \\ V_{oi} = V_n - n_i Q_i \end{cases} \quad (3)$$

where ω_i denotes the frequency reference for the i -th DG's output capacitor, ω_n and V_n are the nominal frequency and voltage, and m_i and n_i are the active and reactive droop coefficients, respectively.

To prevent potential overloads and enhance reliability, it is desirable for all controllable DGs to share the total load power proportionally according to their rated capacities:

$$\begin{cases} \frac{P_1}{P_{n,1}} = \frac{P_2}{P_{n,2}} = \dots = \frac{P_i}{P_{n,i}} \\ \frac{Q_1}{Q_{n,1}} = \frac{Q_2}{Q_{n,2}} = \dots = \frac{Q_i}{Q_{n,i}} \end{cases} \quad (4)$$

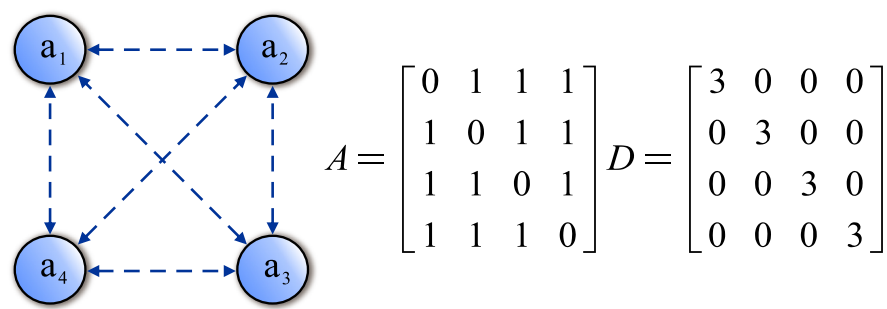


FIGURE 2
Communication graph of multi-agent system.

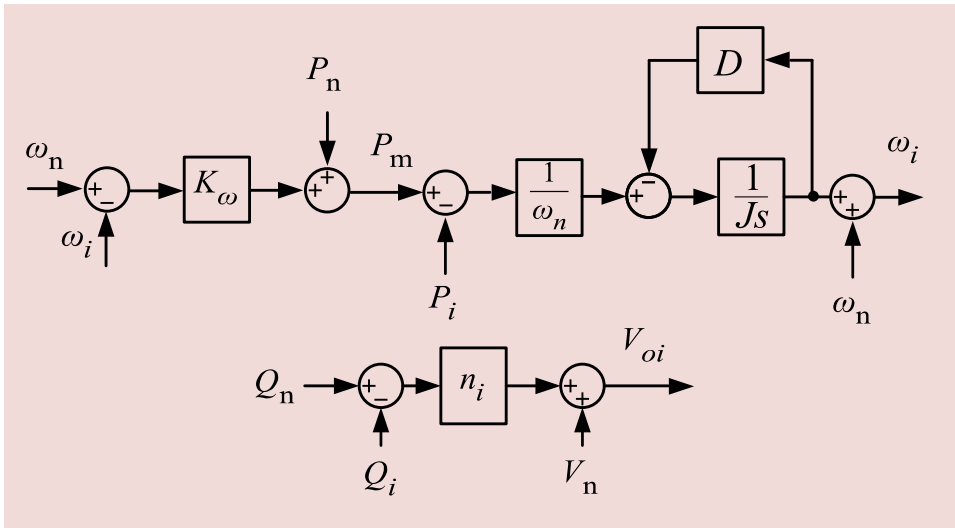


FIGURE 3
VSG control block diagram.

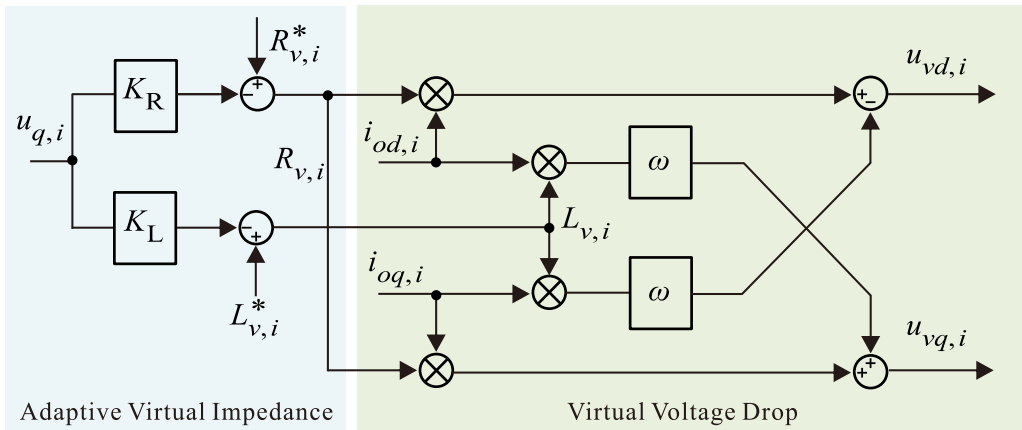


FIGURE 4
Block diagram of AVI generation and control.

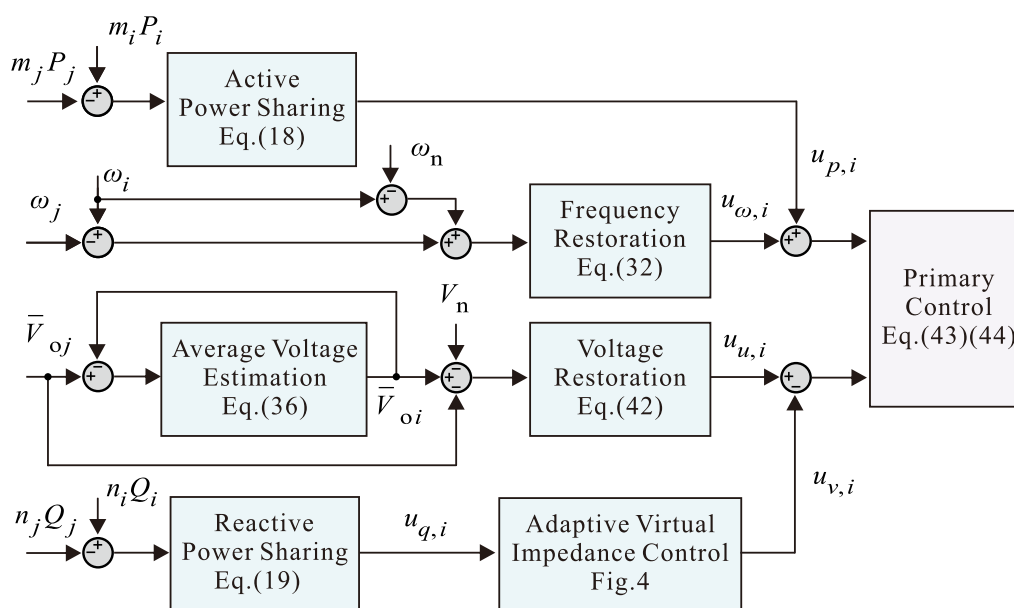


FIGURE 5
Principle of SC structure.

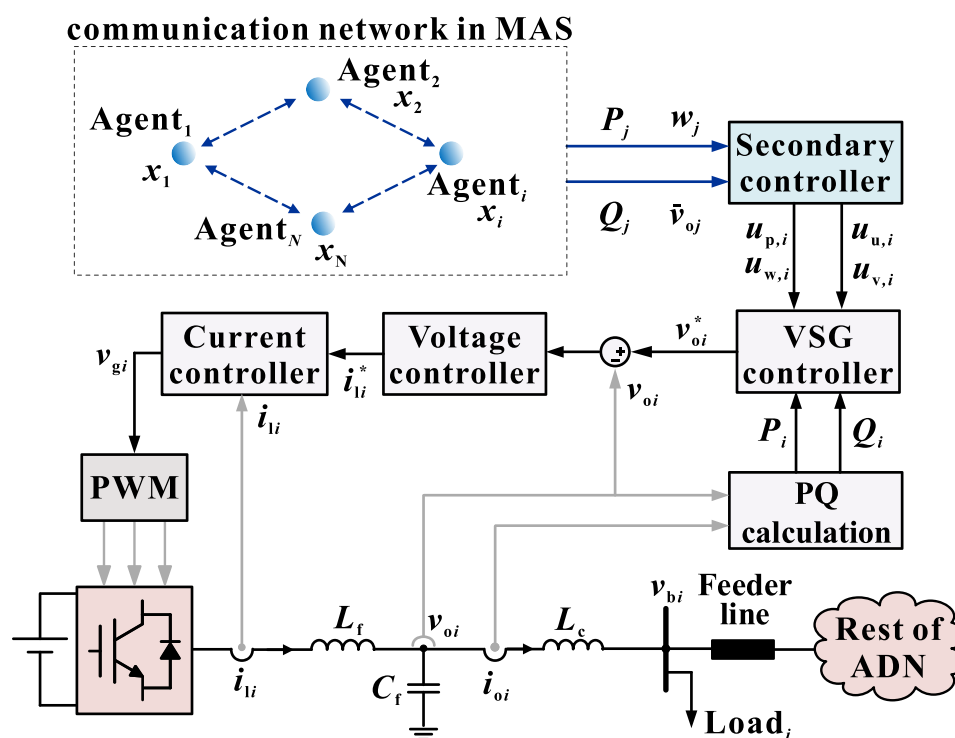


FIGURE 6
Hierarchical control framework in the ADN.

where $P_{n,i}$ and $Q_{n,i}$ are the rated active and reactive power of the i -th DG.

Since frequency is a global variable, DGs synchronize to a common frequency at steady state. Therefore, accurate active power

sharing can be achieved by setting the active droop coefficients inversely proportional to their rated active power:

$$m_1 P_{n,1} = m_2 P_{n,2} = \dots = m_i P_{n,i}. \quad (5)$$

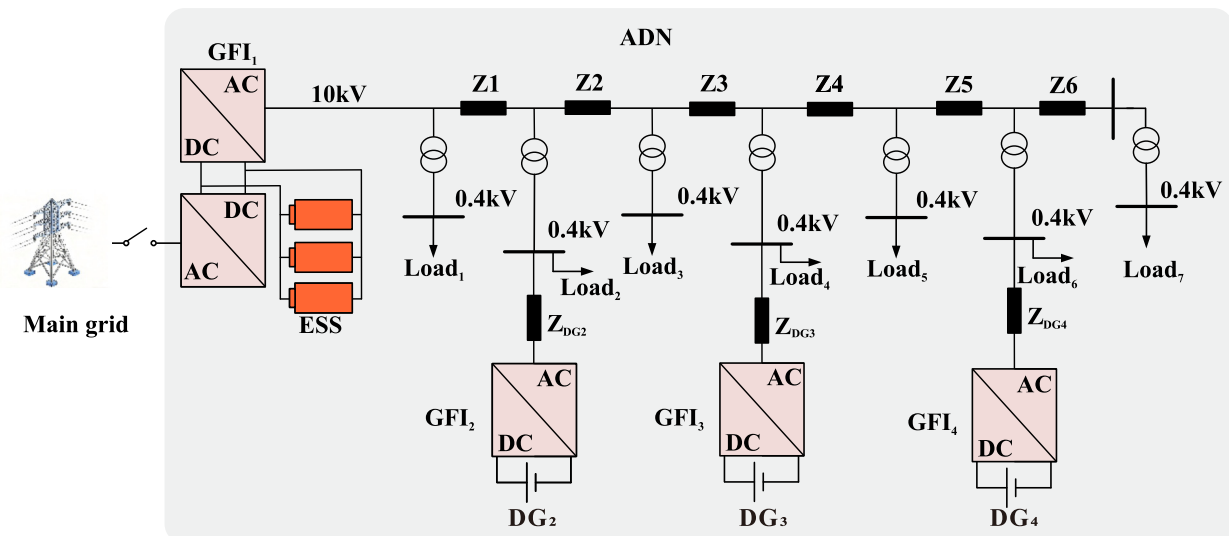


FIGURE 7
The test system of ADN.

However, accurate reactive power sharing is more challenging due to mismatched feeder impedances. Combining Equations 2, 3 yields:

$$n_i Q_i = \frac{(V_{oi} - V_{pcc}) V_{pcc}}{\frac{X_i}{n_i} + V_{pcc}}. \quad (6)$$

Analysis of Equation 6 indicates that for precise reactive power sharing, not only must the reactive droop coefficients satisfy a condition similar to Equation 5:

$$n_1 Q_{n,1} = n_2 Q_{n,2} = \dots = n_i Q_{n,i}, \quad (7)$$

but the feeder reactances must also adhere to a specific proportional relationship:

$$n_1^{-1} X_1 = n_2^{-1} X_2 = \dots = n_i^{-1} X_i. \quad (8)$$

In practice, the conditions of Equations 7, 8 are rarely met due to unequal electrical distances and the resistive nature of low-voltage distribution lines, which also leads to active-reactive power coupling and potential instability issues.

To address these problems, a virtual impedance loop is introduced into the control strategy. It consists of a static virtual inductance and an adaptive virtual impedance. The former ensures the equivalent output impedance is inductive to decouple power flow, while the latter dynamically adjusts to achieve the desired reactive power sharing. The detailed design is presented in Section 3.

$$X'_{li} = X_{li} + X_{v,i} \quad (9)$$

$$X_{vi} = X_{vi}^* + \Delta X_{v,i} \quad (10)$$

where X_{li} is the actual line reactance, $X_{v,i}$ is the total virtual reactance, $X_{v,i}^*$ is the static virtual reactance, and $\Delta X_{v,i}$ is the adaptive virtual reactance component.

2.2 Dynamic consensus algorithm-based multi-agent system

To achieve accurate power sharing while restoring system frequency and average voltage, the DCA-based MAS is widely employed in SC. The communication topology of the MAS is generally described by a weighted directed graph $G = (V_G, E_G, A_G)$. Here, $V_G = \{v_1, v_2, \dots, v_N\}$ is the set of nodes, each corresponding to a DG agent. The edge set $E_G \subseteq V_G \times V_G$ represents communication links between nodes. The adjacency matrix $A = [a_{ij}] \in \mathbb{R}^{N \times N}$ and the in-degree matrix $D = [d_{ii}] \in \mathbb{R}^{N \times N}$ define the interconnection:

$$a_{ij} = \begin{cases} 1, & (v_i, v_j) \in E_G \\ 0, & (v_i, v_j) \notin E_G \end{cases} \quad (11)$$

$$d_{ii} = \begin{cases} \sum_{j=1}^N a_{ij}, & i = j \\ 0, & i \neq j \end{cases} \quad (12)$$

where $a_{ij} = 1$ indicates that agents i and j are communicated; d_{ii} is the total number of agents communicated with agent i . The system Laplacian matrix is defined as $L = D - A$. Figure 2 illustrates an example communication topology with four agents and its corresponding matrices.

Under such a topology, each agent updates its state by exchanging information with the linked neighbors according to the DCA to achieve consensus. The basic continuous-time consensus protocol is:

$$\dot{x}_i(t) = - \sum_{j \in N_i} a_{ij} (x_i(t) - x_j(t)), \quad (13)$$

where $x_i(t)$ is the state of agent i and N_i is its linked neighbor set.

TABLE 1 Electrical and control parameters.

Parameter	Symbol	Value
Line voltage base (V)	V_{base}	400
Apparent power base (kVA)	S_{base}	100
LC filter inductance (mH)	L_f	0.3
LC filter capacitance (μF)	C_f	0.012
Line impedance (Ω)	Z_{1-6}	$1.5 + j1.75$
DG1 line impedance (Ω)	Z_{DG1}	0
DG2 line impedance (Ω)	Z_{DG2}	$0.1 + j0.2$
DG3 line impedance (Ω)	Z_{DG3}	$0.3 + j0.4$
DG4 line impedance (Ω)	Z_{DG4}	$0.15 + j0.25$
Rated frequency (rad/s)	ω_0	314
Speed regulation coefficient	K_ω	10
Virtual moment of inertia ($\text{kg}\cdot\text{m}^2$)	J_{1-4}	0.5
Damping coefficient of in DG1 (N·m)	D_1	15
Damping coefficient of DG2 (N·m)	D_2	22.5
Damping coefficient of DG3 (N·m)	D_3	30
Damping coefficient of DG4 (N·m)	D_4	37.5
Reactive power droop coefficient of DG1	n_1	0.0006
Reactive power droop coefficient of DG2	n_2	0.0004
Reactive power droop coefficient of DG3	n_3	0.0003
Reactive power droop coefficient of DG4	n_4	0.00024
Frequency compensation gain	$c_{\omega,1-4}$	30
Active power compensation gain	$c_{p,1-4}$	30
Voltage compensation gain	$c_{u,1-4}$	10
Reactive power compensation gain	$c_{q,1-4}$	2
Virtual resistance gain	K_R	0.001
Virtual inductance gain	K_L	0.0001

To enable convergence to a desired reference value x_{ref} , a reference feedback term is incorporated:

$$\dot{x}_i(t) = - \sum_{j \in N_i} [a_{ij}(x_i(t) - x_j(t)) + g_i(x_i - x_{\text{ref}}(t))] \quad (14)$$

where $g_i \geq 0$ is the tracking gain. If at least one agent has $g_i > 0$, all agents' states will asymptotically converge to x_{ref} .

3 Construction of the distributed hierarchical control strategy

3.1 Primary control based on local measurements

Due to the fast response, droop control is widely used in PC. However, it lacks inertia and damping. To balance power fluctuations and provide necessary support capability for the

grid, this paper employs Virtual Synchronous Generator (VSG) control Babayomi et al. (2020). Its core lies in using a control algorithm to make the grid-connected inverter emulate the electromechanical transient process of a synchronous generator's rotor, as shown in Figure 3. The physical principles of this control strategy can be described as:

$$\begin{cases} J \frac{d\Delta\omega}{dt} = \frac{P_m}{\omega_n} - \frac{P_i}{\omega_n} - D(\omega_i - \omega_n) \\ P_m = P_n + K_\omega(\omega_n - \omega_i) \end{cases} \quad (15)$$

where ω_n and ω_i are the rated and actual rotor angular frequencies, respectively; J and D are the virtual inertia and damping coefficient, respectively; K_ω is the speed regulation coefficient; P_m is the mechanical power.

Considering $P_n = 0$ (15) can be transformed into that,

$$\omega_i = \omega_n - \frac{m_i}{\tau s + 1} P_i, \quad (16)$$

$$\begin{cases} \tau = \frac{J\omega_n}{D\omega_n + K_\omega}, \\ m_i = \frac{1}{D\omega_n + K_\omega}, \end{cases} \quad (17)$$

where τ is the first-order inertial time constant, and m_i is the equivalent active power-frequency droop coefficient.

Since the VSG control is equivalent to droop control in steady state, the theoretical analysis of power sharing in Section 2 remains applicable Meng et al. (2019). The PC can quickly respond to frequency and voltage changes. It autonomously regulates active and reactive power output based solely on local measurements, initially establishing the system's frequency and voltage foundation and maintaining instantaneous system balance.

3.2 Secondary control based on sparse communication network

3.2.1 Power sharing control

As indicated by 2 and 8, P_i can be regulated through δ_p , while the accurate sharing of Q_i relies on the reshaping of feeder impedance. To achieve accurate proportional power sharing, it is necessary to employ consensus protocols to drive the $m_i P_i$ and $n_i Q_i$ values of different DGs towards consensus. For active power, a leaderless consensus algorithm can be used to correct the frequency reference. In contrast, accurate reactive power sharing is more complex. This paper employs adaptive virtual impedance control to compensate for mismatched feeder impedance.

$$u_{p,i} = -c_p \int \sum_{j \in N_i} a_{ij} (m_i P_i - m_j P_j) dt \quad (18)$$

$$u_{q,i} = -c_q \int \sum_{j \in N_i} a_{ij} (n_i Q_i - n_j Q_j) dt \quad (19)$$

where c_p and c_q are the active and reactive compensation gains, respectively (both positive), and $u_{p,i}$ and $u_{q,i}$ are the compensation values required to achieve active and reactive power sharing, respectively.

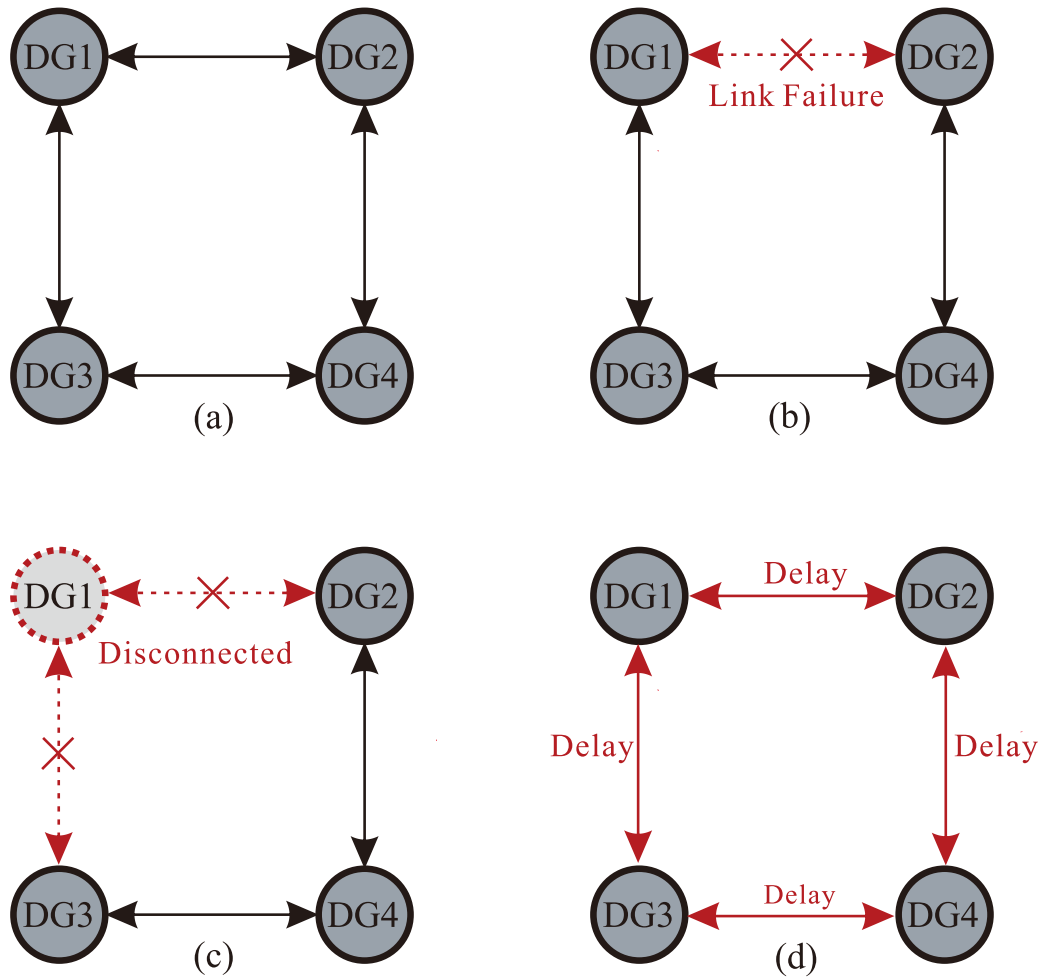


FIGURE 8
The illustration of different cases. (a) case 1. (b) case 2. (c) case 3. (d) case 4.

Therefore, the output power change rate of a DG can be regulated by the compensation value. Taking active power as an example:

$$m_i \dot{P}_i = \dot{u}_{p,i} \quad (20)$$

According to the consensus protocol, the compensation value $\dot{u}_{p,i}$ for active power sharing is derived from the DG's own information and its neighbors' information. From Equation 18:

$$\dot{u}_{p,i} = -c_p e_{p,i} \quad (21)$$

where $e_{p,i}$ is the active power sharing error between the local DG and its neighbors:

$$e_{p,i} = \sum_{j \in N_i} a_{ij} (m_i P_i - m_j P_j) \quad (22)$$

Expressing the entire system in matrix form yields:

$$m \dot{P} = \dot{u}_p \quad (23)$$

$$\dot{u}_p = -c_p e_p \quad (24)$$

Based on the definitions of the global variables $mP = [m_1 P_1, m_2 P_2, \dots, m_N P_N]^T$, $\dot{u}_p = [\dot{u}_{p,1}, \dot{u}_{p,2}, \dots, \dot{u}_{p,N}]^T$, and $e_p = [e_{p,1}, e_{p,2}, \dots, e_{p,N}]^T$, Equation 22 can be expressed in a compact matrix form:

$$e_p = LmP \quad (25)$$

To analyze the stability, consider the following Lyapunov function candidate:

$$V_p = \frac{1}{2} e_p^T R e_p \quad (26)$$

where $R = R^T$ is a positive definite matrix. Taking the derivative of Equation 26 yields:

$$\dot{V}_p = e_p^T R \dot{e}_p = e_p^T R L m \dot{P} \quad (27)$$

Substituting Equations 23, 24 into Equation 27 gives:

$$\dot{V}_p = -c_p e_p^T R L e_p \quad (28)$$

Since V_p is a scalar, $V_p = (V_p)^T$, therefore,

$$\begin{aligned} \dot{V}_p &= \frac{1}{2} [\dot{V}_p + (\dot{V}_p)^T] \\ &= -\frac{1}{2} c_p e_p^T (RL + L^T R) e_p \end{aligned} \quad (29)$$

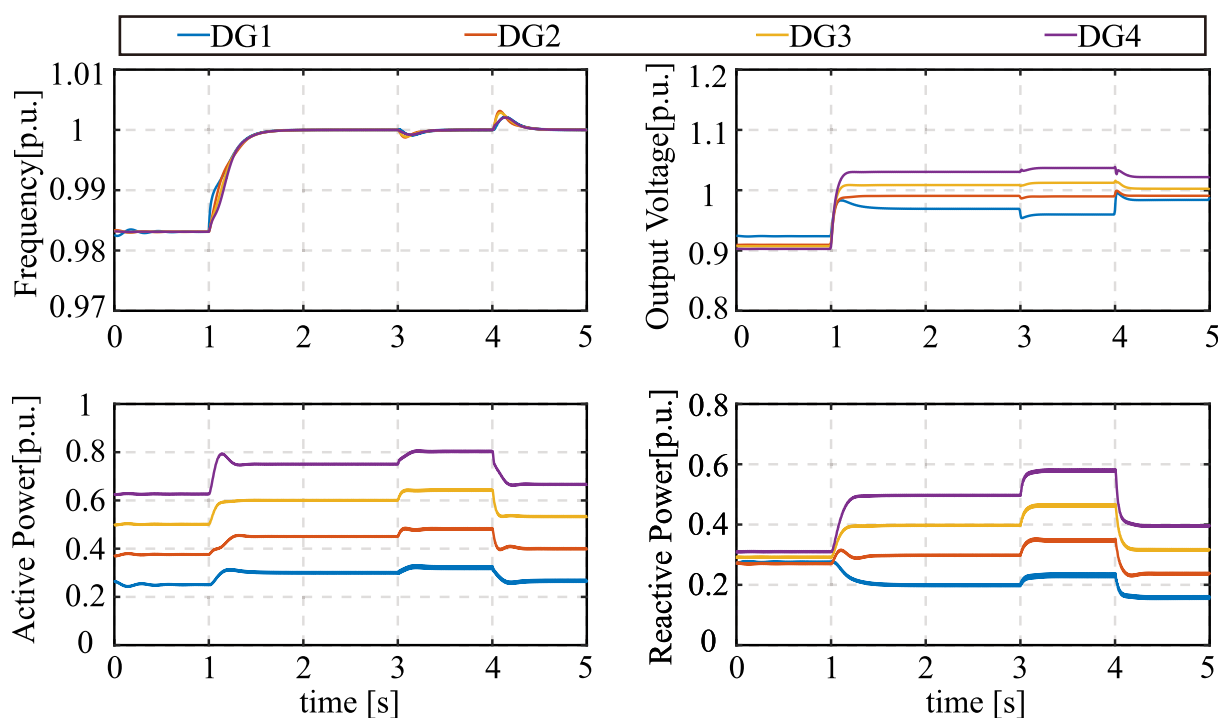


FIGURE 9 $\omega/V/P/Q$ time-domain performance under load change with intact communication.

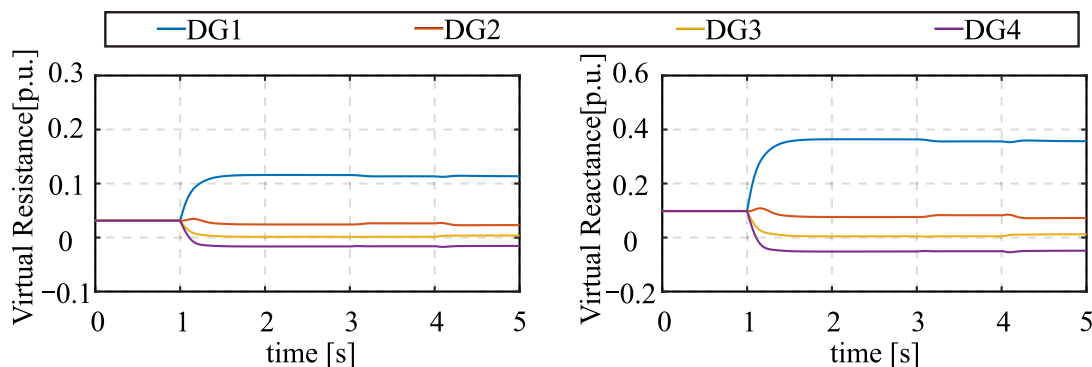


FIGURE 10 Variation of virtual impedance compensation.

According to Olfati-Saber and Murray (2004), the matrix $(\mathbf{RL} + \mathbf{L}^T \mathbf{R})$ is positive definite. Consequently, $\dot{V}_p < 0$, proving that the proposed power sharing control is asymptotically stable, and the active power sharing error will converge to zero.

It is important to note that the above stability analysis also applies to reactive power sharing. However, unlike the active power compensation $u_{p,i}$ which directly adjusts the frequency reference, the reactive power compensation term, $u_{q,i}$, is utilized to generate an adaptive virtual impedance. This addresses the feeder impedance mismatch problem discussed in Section 2, thereby enhancing system performance. As shown in Figure 4 and implemented in the dq -axis

frame, this is described by:

$$\begin{cases} L_{v,i} = L_{v,i}^* - K_L u_{q,i}, \\ R_{v,i} = R_{v,i}^* - K_R u_{q,i}, \end{cases} \quad (30)$$

where $L_{v,i}^*$ and $R_{v,i}^*$ are the static virtual impedance components, K_L and K_R are scalar gains, and $L_{v,i}$ and $R_{v,i}$ are the resulting adaptive virtual impedance.

It should be noted that the main purpose of $L_{v,i}$ is to rapidly compensate for line reactance mismatch in order to achieve accurate reactive power sharing. If $L_{v,i}$ is too small, the virtual inductance adjustment becomes sluggish, resulting in slow system convergence

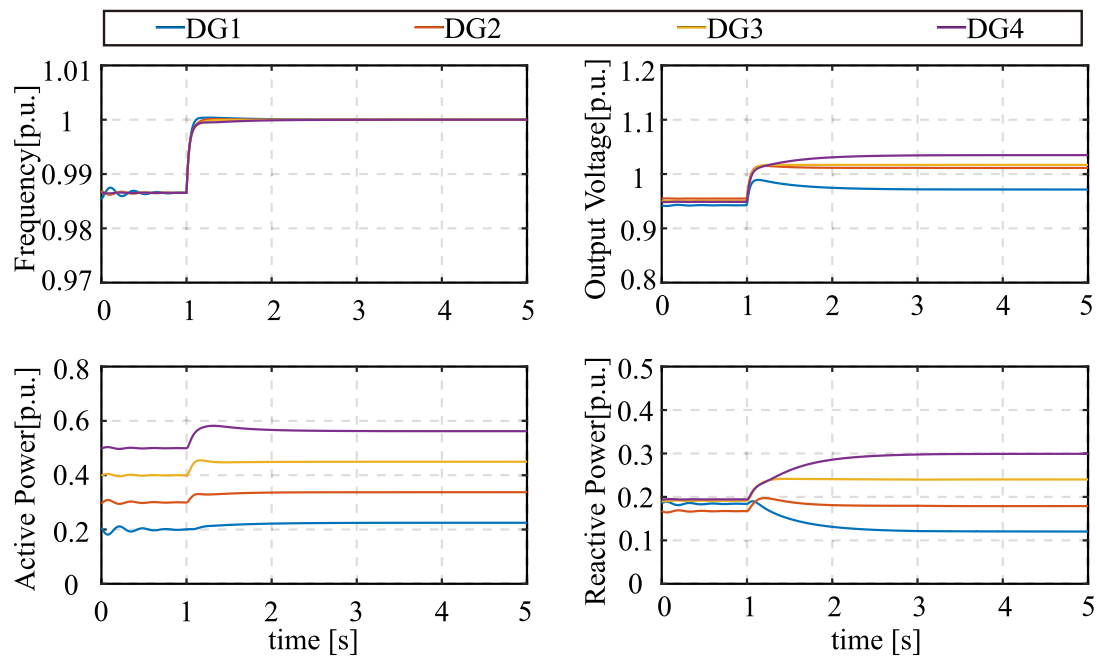


FIGURE 11 $\omega/V/P/Q$ time-domain performance under partial communication failure.

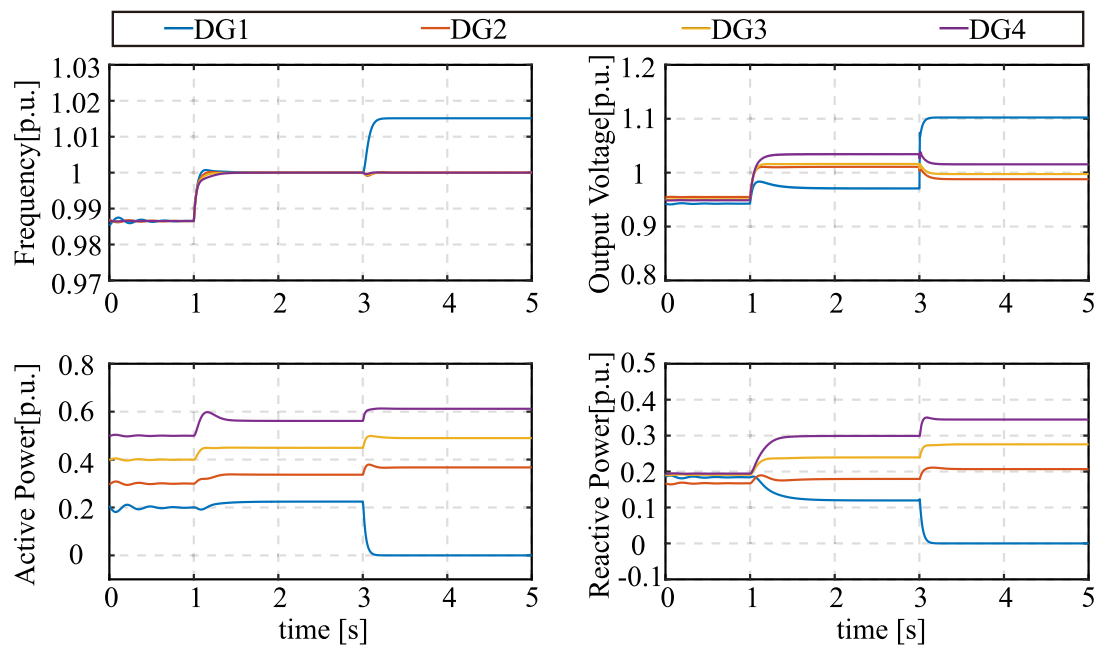


FIGURE 12 $\omega/V/P/Q$ time-domain performance upon DG1 Outage.

and poor disturbance rejection. Conversely, an excessively large $L_{v,i}$ may lead to excessive sensitivity, potentially causing oscillations or overshoot. In cases where feeder impedances differ substantially, a larger K_L can be adopted to accelerate the compensation of line impedance mismatch. Meanwhile, $R_{v,i}$ provides damping to the

system, helping to suppress circulating currents and oscillations among units. To maintain a predominantly inductive equivalent impedance, which is beneficial for P - Q decoupling, K_R is generally kept small to avoid introducing significant voltage drops and additional active power losses. Based on the above considerations

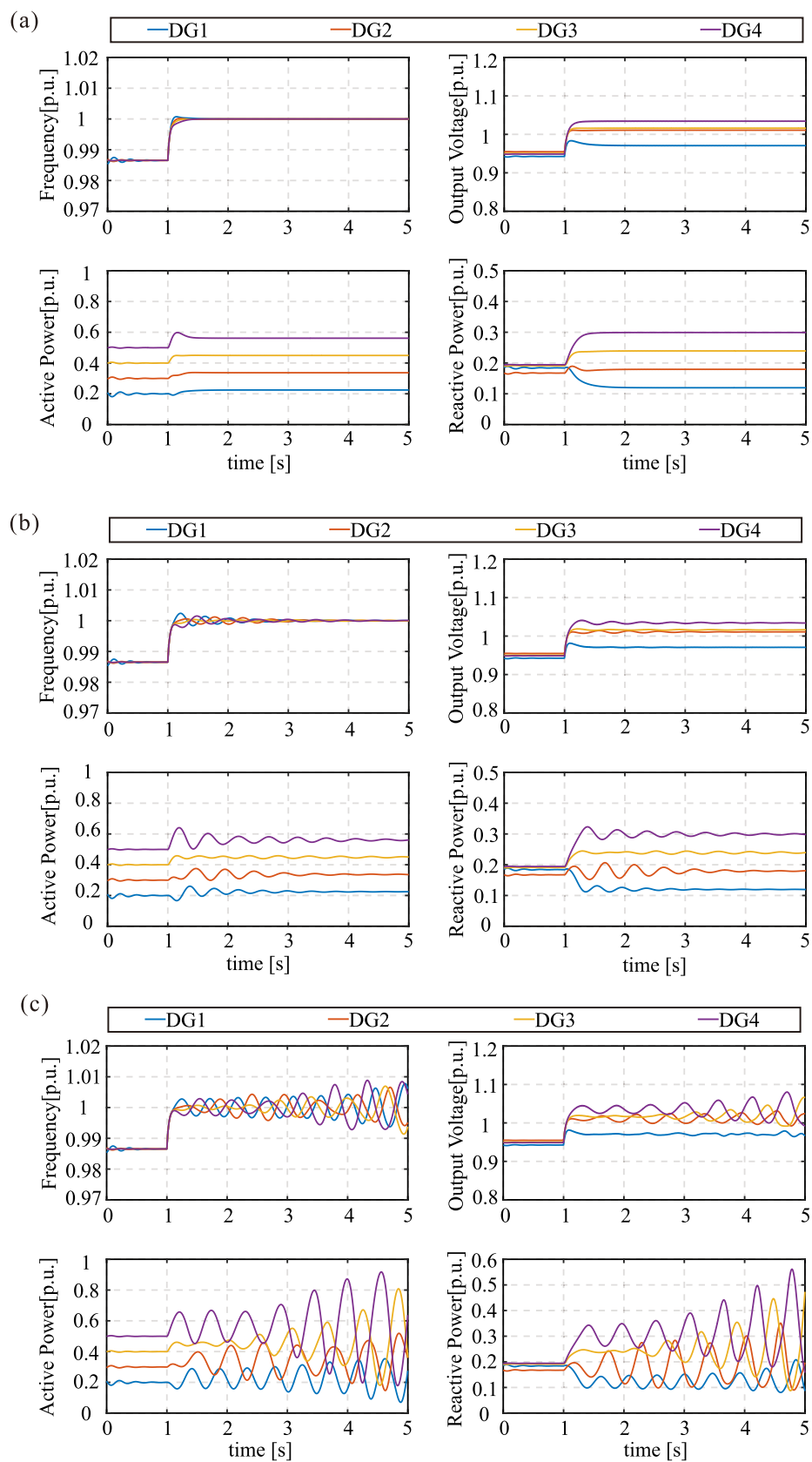


FIGURE 13 $\omega/V/P/Q$ time-domain performance under different communication delays. (a) delay = 200 ms. (b) delay = 250 ms. (c) delay = 300 ms.

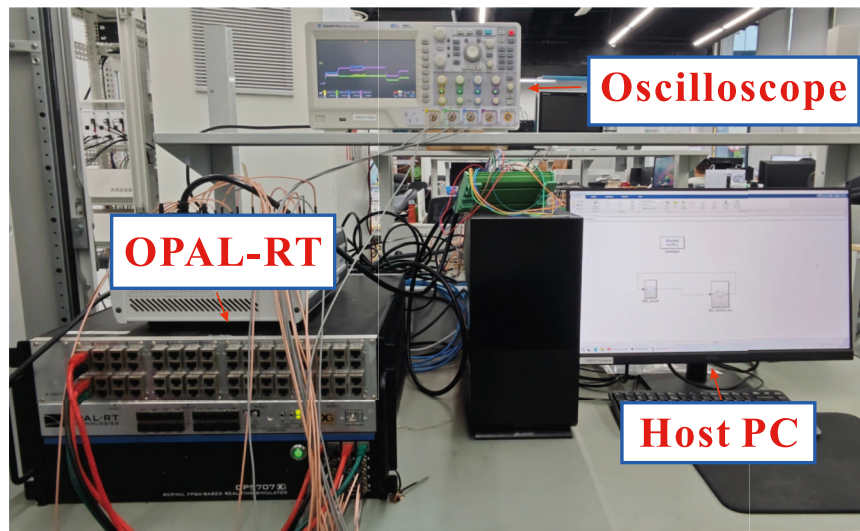


FIGURE 14
OPAL-RT real-time simulation platform.

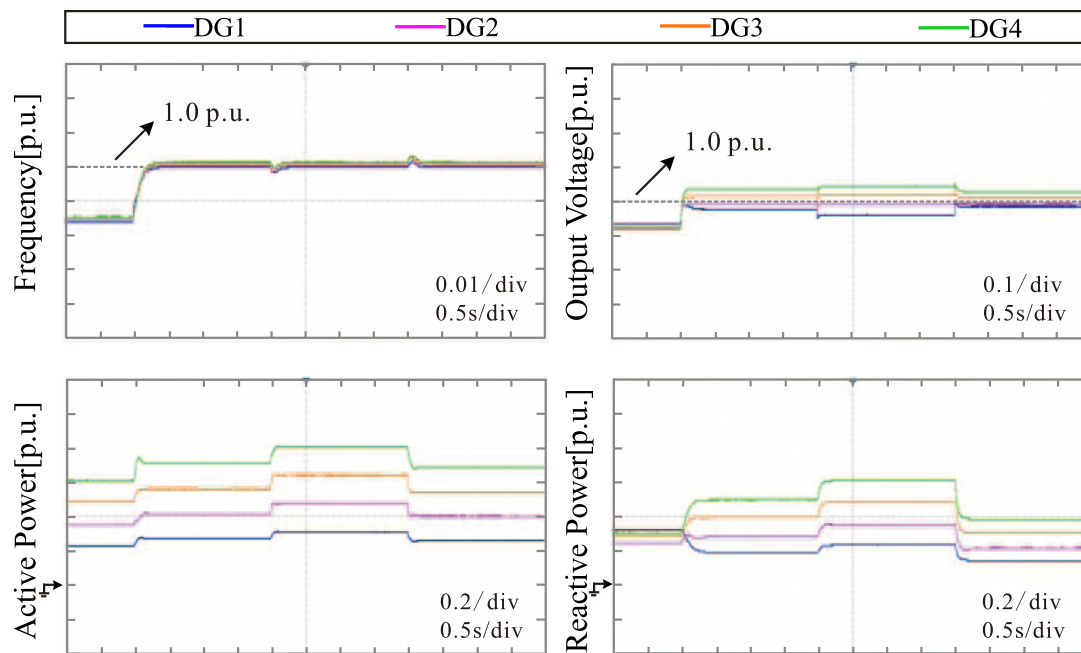


FIGURE 15
Performance of the system in case 1.

and thorough parameter tuning, the values chosen for the case studies in this paper are $K_L = 0.0001$ and $K_R = 0.001$.

The virtual impedance is realized by introducing a virtual voltage drop:

$$\begin{cases} u_{vd,i} = i_{od,i}R_{v,i} - i_{oq,i}\omega L_{v,i}, \\ u_{vq,i} = i_{oq,i}R_{v,i} + i_{od,i}\omega L_{v,i}, \end{cases} \quad (31)$$

where $i_{od,i}$ and $i_{oq,i}$ represent the output currents of the i -th DG, and $u_{vd,i}$ and $u_{vq,i}$ are the virtual voltage drops

induced by the virtual impedance. When a reactive power sharing error exists, the equivalent feeder impedance is continuously adjusted until the desired sharing objective is achieved.

3.2.2 Frequency and voltage restoration control

When the actual output power of the inverters does not match the power commands, the system frequency and voltage deviate from their rated values. Therefore, the SC

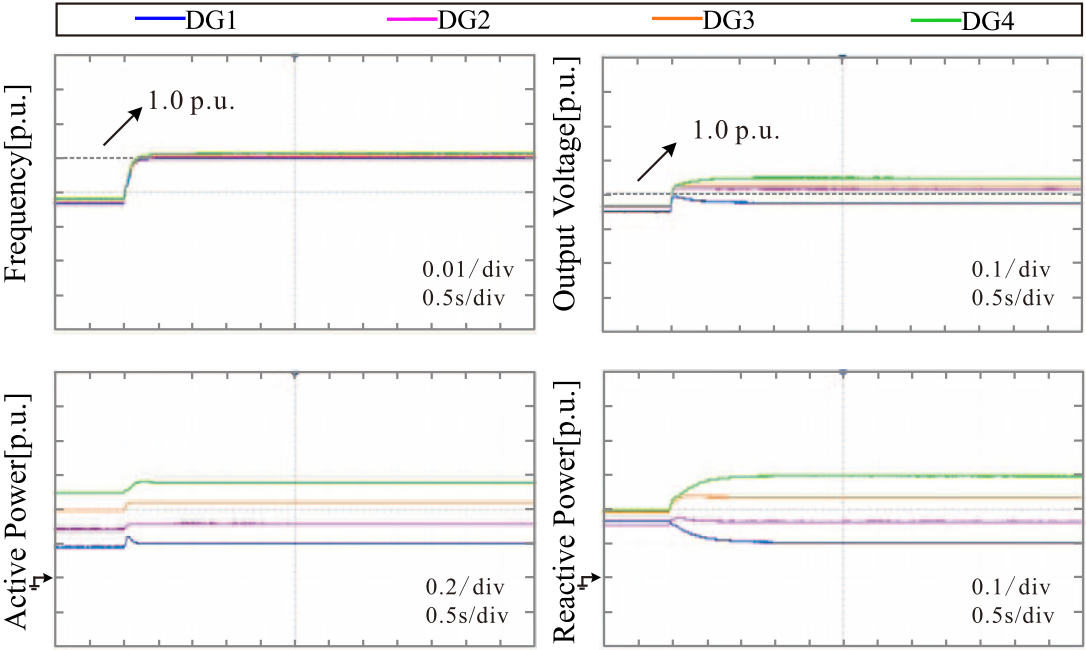


FIGURE 16
Performance of the system in case 2.

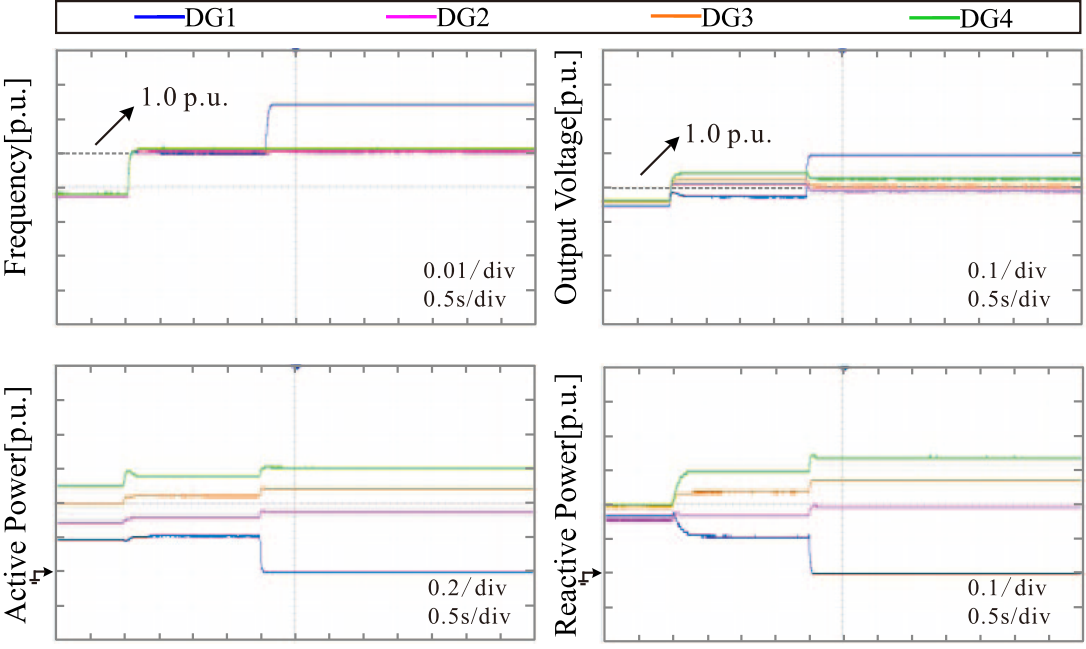
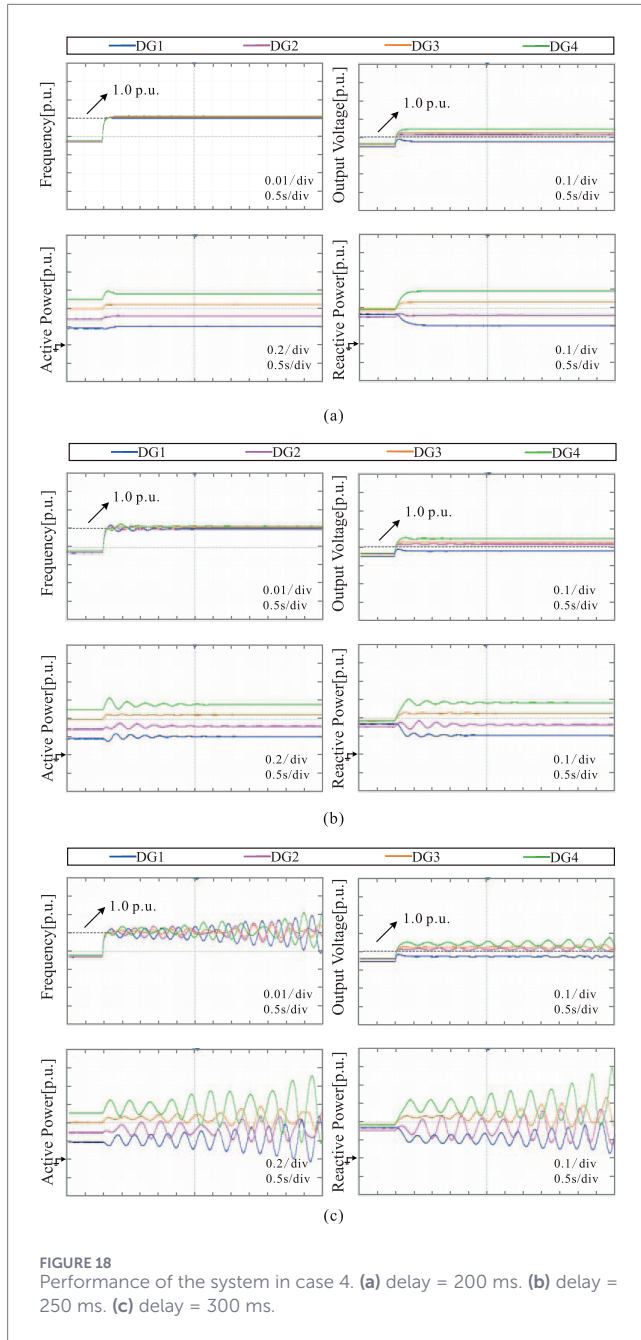


FIGURE 17
Performance of the system in case 3.



employs a leader-following consensus algorithm to restore them.

For frequency restoration, the control law is:

$$u_{\omega,i} = -c_{\omega} \int \sum_{j \in N_i} a_{ij} ((\omega_i - \omega_j) + g_i (\omega_i - \omega_n)) dt \quad (32)$$

where c_{ω} is the frequency compensation gain, and $u_{\omega,i}$ is the frequency restoration compensation value.

Defining the state error as $e_{\omega,i} = \omega_i - \omega_n$ and considering only the frequency dynamics $\dot{\omega}_i = \dot{u}_{\omega,i}$, the rate of change of the frequency error is:

$$\dot{e}_{\omega,i} = -c_{\omega} \sum_{j \in N_i} a_{ij} ((\omega_i - \omega_n) - (\omega_j - \omega_n) + g_i e_{\omega,i}) \quad (33)$$

The global frequency error dynamics for the entire system can be described as:

$$\dot{e}_{\omega}(t) = -c_{\omega} (L + G) e_{\omega}(t) \quad (34)$$

where $G = \text{diag}(g_1, g_2, \dots, g_N)$. The matrix $-c_{\omega}(L + G)$, related to the algebraic connectivity of the augmented communication graph, determines the convergence speed. The solution to Equation 34 is:

$$e_{\omega}(t) = e^{-c_{\omega}(L+G)t} e_{\omega}(t_0) \quad (35)$$

According to Zhou et al. (2020), $-c_{\omega}(L + G)$ is negative definite and invertible, ensuring that the frequency error $e_{\omega}(t)$ gradually converges to zero.

For voltage restoration, since voltage differences across nodes are inevitable due to feeder impedances, this paper adopts an average voltage restoration scheme. A distributed average voltage observer is designed as:

$$\dot{\bar{V}}_{oi}(t) = V_{oi}(t) + c_e \int \sum_{j \in N_i} a_{ij} (\bar{V}_{oj}(t) - \bar{V}_{oi}(t)) dt, \quad (36)$$

where $V_{oi}(t)$ is the actual output voltage of the i -th inverter, $\bar{V}_{oi}(t)$ is its estimated average voltage, and c_e is the coupling gain. The observer dynamics are:

$$\dot{\bar{V}}_{oi}(t) = \dot{V}_{oi}(t) + c_e \sum_{j \in N_i} a_{ij} (\bar{V}_{oj}(t) - \bar{V}_{oi}(t)) \quad (37)$$

The global dynamics of the distributed voltage observer can be described as:

$$\dot{\bar{V}}_o(t) = \dot{V}_o(t) - c_e L \bar{V}_o(t) \quad (38)$$

where $\bar{V}_o(t) = [\bar{V}_{o1}(t), \bar{V}_{o2}(t), \dots, \bar{V}_{oN}(t)]^T$ and $V_o(t) = [V_{o1}(t), V_{o2}(t), \dots, V_{oN}(t)]^T$. Assuming zero initial conditions and applying the Laplace transform to Equation 38, we obtain:

$$\bar{V}_o(s) = s(sI + c_e L)^{-1} V_o(s) \quad (39)$$

Applying the final value theorem:

$$\lim_{t \rightarrow \infty} \bar{V}_o(t) = \lim_{s \rightarrow 0} s \cdot [s(sI + c_e L)^{-1} V_o(s)] \quad (40)$$

According to Ning et al. (2021), $\lim_{s \rightarrow 0} s(sI + c_e L)^{-1} = \frac{1}{N} \mathbf{1}\mathbf{1}^T$, where $\mathbf{1}$ is a vector of all ones. Substituting this into Equation 40 yields:

$$\begin{aligned} \lim_{t \rightarrow \infty} \bar{V}_o(t) &= \left(\frac{1}{N} \mathbf{1}\mathbf{1}^T \right) \cdot \lim_{s \rightarrow 0} s V_o(s) \\ &= \left(\frac{1}{N} \mathbf{1}\mathbf{1}^T \right) \cdot \lim_{t \rightarrow \infty} V_o(t) \\ &= \left(\frac{1}{N} \mathbf{1}\mathbf{1}^T \right) V_o^{ss} \end{aligned} \quad (41)$$

where V_o^{ss} is the steady-state DG output voltage vector.

Therefore, through this distributed observer, each DG can obtain the global average system voltage. Subsequently, similar to frequency restoration, a leader-following consensus algorithm is used to restore the average voltage to its rated value:

$$u_{u,i} = -c_u \int \sum_{j \in N_i} a_{ij} ((\bar{V}_{oi} - \bar{V}_{oj}) + g_i (\bar{V}_{oi} - V_n)) dt \quad (42)$$

where c_u is the voltage compensation gain, and $u_{u,i}$ is the voltage restoration compensation value.

3.3 Integrated control structure and objectives

As illustrated in Figure 5, unlike the PC, the SC relies on a sparse communication network to exchange information with neighboring DGs. It employs the DCA to adjust the reference values for the PC, achieving accurate power sharing and frequency/voltage restoration.

The complete distributed hierarchical control strategy is depicted in Figure 6. The frequency and voltage reference values, after correction by the SC, are described by:

$$\omega_i^* = \omega_n - m_i P_i + u_{p,i} + u_{\omega,i} \quad (43)$$

$$\begin{cases} V_{di}^* = V_n - n_i Q_i - u_{vd,i} + u_{u,i} \\ V_{qi}^* = 0 - u_{vq,i} \end{cases} \quad (44)$$

For a system comprising N DGs, accurate power sharing alongside frequency and voltage restoration is achieved, satisfying the following control objectives:

$$\begin{cases} \lim_{t \rightarrow \infty} \omega_i(t) = \omega_n, \quad \forall i \in \{1, 2, \dots, N\} \\ \lim_{t \rightarrow \infty} \bar{V}_{oi}(t) = V_n, \quad \forall i \in \{1, 2, \dots, N\} \end{cases} \quad (45)$$

$$\begin{cases} \lim_{t \rightarrow \infty} |m_i P_i(t) - m_j P_j(t)| = 0, \quad \forall i, j \in \{1, 2, \dots, N\} \\ \lim_{t \rightarrow \infty} |n_i Q_i(t) - n_j Q_j(t)| = 0, \quad \forall i, j \in \{1, 2, \dots, N\} \end{cases} \quad (46)$$

4 Case studies

To validate the proposed control strategy, case studies based on a real low-voltage distribution network model from the Xinjiang region in China are tested and analyzed in MATLAB/Simulink, with a simulation step size of 1×10^{-6} s. When a severe fault occurs in the main grid, the distribution network disconnects from the upstream grid, and four inverter-based DGs or ESS coordinately supply power to critical loads. The system structure is shown in Figure 7, with detailed parameters listed in Table 1. Four different test scenarios are designed, as illustrated in Figure 8, and the results are presented in per-unit values.

4.1 Case 1: general performance evaluation

The test scenario is shown in Figure 8a, which evaluates the system's power-sharing capability under normal communication conditions. The capacity ratio of the four DGs is assumed to be 1:1.5:2:2.5, and their equivalent droop coefficients have been set inversely proportional to their capacities (see Table 1). Under the proposed HC, the expected power sharing ratio should match the capacity ratio. As shown in Figure 9, from 0 to 1 s, the system establishes preliminary voltage and frequency levels under PC, and a simple power sharing is achieved. However, due to feeder impedance mismatches, accurate power sharing is not realized, and the system frequency and voltage experience a drop. At $t = 1$ s, the SC is activated. Through the communication network-based consensus algorithm, as depicted in Figure 10, virtual impedance is generated

for compensation, and frequency/voltage restoration is performed. Under the proposed HC, by $t = 2$ s, the expected power sharing ratio is achieved. To simulate actual load fluctuations, an additional load of 100 kW + 30 kVar is connected at $t = 3$ s, and a load of 150 kW + 50 kVar is disconnected at $t = 4$ s. The test results demonstrate that the system maintains proportional power sharing with minimal frequency and voltage fluctuations.

4.2 Case 2: communication link failure

The test scenario is shown in Figure 8b, which investigates the impact of communication link failure between some DGs. At $t = 2$ s, coinciding with the activation of the SC, the communication link between DG1 and DG2 fails. As can be seen from Figure 11, although the system's convergence time becomes longer, the power sharing capability is not affected, and both frequency and voltage are restored to their nominal ranges. This is because the strong connectivity of the communication graph is not destroyed despite the partial link failure, and system synchronization can still be maintained under the action of the consensus algorithm.

4.3 Case 3: plug-and-play capability

To test the system's "plug-and-play" capability, as shown in Figure 8b, DG1 is assumed to fail and disconnect from the system at $t = 3$ s, which also means its communication links with DG2 and DG3 are lost. The results in Figure 12 show that after DG1 disconnects, its output power slowly decays to zero due to the output filter. Meanwhile, the remaining DGs continue to share the load proportionally according to their capacity ratio of 1.5:2:2.5, with only minor frequency and voltage fluctuations. It is worth noting that after DG1 exits the system, the correction values for its voltage and frequency generated by the SC are no longer updated. As its output power decreases, its frequency and voltage will significantly deviate from the nominal values due to the droop characteristic, but this undoubtedly has no impact on the main system as it is already disconnected.

4.4 Case 4: communication latency

The test scenario is shown in Figure 8d. This case primarily investigates the impact of communication latency on the proposed control strategy. At $t = 1$ s, coinciding with the activation of the SC, communication delays of different magnitudes (200 ms, 250 ms, and 300 ms) are introduced to assess their effects on system performance. As can be observed from Figure 13, the system can tolerate a communication latency of up to 250 ms, maintaining stable operation and achieving the control objectives. However, as the delay increases, system performance degrades. With a 250 ms delay, noticeable oscillations appear in the dynamic response. Furthermore, with a 300 ms delay, the system becomes unstable. These results indicate that, while the proposed distributed control strategy exhibits a certain degree of resilience to communication delays, its performance is constrained by the delay margin inherent to the consensus algorithm and the system's dynamic characteristics.

5 Hardware-in-the-loop verification based on Opal-RT platform

To more closely approximate real-world conditions and further validate the theoretical analysis and simulation results presented in the preceding sections, this work employs the OPAL-RT LAB 5707 real-time digital simulator to construct a hardware-in-the-loop platform. The overall setup of this platform is depicted in Figure 14.

Experimental tests corresponding to the simulation cases in Section 4 were conducted, with the results presented in Figures 15–18. Figure 15 demonstrates the system performance under normal HC, featuring accurate proportional power sharing and frequency/voltage restoration. Under the action of the PC alone, active power sharing is satisfactory; however, reactive power is nearly equally shared, deviating from the expected proportional sharing based on capacity ratings. Consequently, the system frequency and voltage drop to approximately 0.982 p. u. and 0.9 p. u., respectively. Upon activation of the SC, the frequency and average voltage are rapidly restored to their rated values and remain stable during subsequent load-switching events. Figure 16 illustrates the system's robustness when subjected to communication-link failures. Consistent with the simulation results, even with partial link interruptions, the system achieves the intended control objectives, albeit with slower convergence as the communication-network connectivity degrades. The experimental results in Figure 17 validate the plug-and-play capability of the proposed strategy: the system maintains stable operation after DG1 is disconnected. At $t = 3$ s, following the disconnection of DG1, the remaining DGs continue to share the system load proportionally according to their respective capacities, experiencing only minor frequency and voltage transients. Finally, Figure 18 demonstrates the system's tolerance to communication delays within a specific margin. The system operates normally when the communication delay is below 250 ms; however, as the delay increases, the system gradually becomes unstable without additional control measures.

6 Conclusion

This paper proposes a hierarchical control strategy for the islanded autonomous operation of active distribution networks. Specifically, the primary control adopts virtual synchronous generator control to provide voltage and frequency support. The secondary control achieves accurate power sharing and frequency/voltage restoration by modifying the reference setpoints via compensation values. In this process, the adaptive virtual impedance technique is embedded within the dynamic consensus algorithm. This approach requires information only from neighboring units, effectively reshaping the equivalent impedance between distributed generators and the point of common coupling to enhance control performance. The proposed method is applicable under unknown grid topology, line impedance, and load conditions. It exhibits a certain tolerance to common communication issues such as link failures and delays and supports the plug-and-play functionality of distributed generators. Finally, the Matlab/Simulink and OPAL-RT 5707 platform based results clearly validate the effectiveness of the proposed strategy.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding author.

Author contributions

XWa: Writing – original draft, Methodology, Supervision, Writing – review and editing. ZZ: Writing – review and editing, Methodology, Conceptualization. YD: Project administration, Visualization, Writing – original draft, Investigation. XWu: Investigation, Writing – original draft, Validation, Methodology. JM: Resources, Project administration, Writing – review and editing, Methodology. XM: Formal Analysis, Resources, Writing – review and editing, Data curation.

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Conflict of interest

Authors XWa, ZZ, YD, XWu, JM, and XiM were employed by Electric Power Research Institute of State Grid Xinjiang Electric Power Co., Ltd.

Generative AI statement

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fenrg.2026.1781946/full#supplementary-material>

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